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Electronic device and user interface display method

Abstract

An electronic device and method are disclosed. The electronic device includes a display an extended screen area that is exposed or stowed, according to a change in a state of the display of the electronic device, communication circuitry configured to transceive a signal to or using an angle-of-arrival (AOA) antenna disposed within the extended screen area, a memory, and a processor. The processor implements the method, including: determining a position of the electronic device relative to an external electronic device, using the communication circuitry, executing a sharing operation with the external electronic device, identifying whether the display of the electronic device is disposed in a first state or a second state, based on detecting that the display is disposed in the first state, perform a first positioning operation for the external electronic device, and displaying a sharing interface according to the first positioning operation on the display.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of U.S. patent application Ser. No. 17/531,871 filed on Nov. 22, 2021, which is a Continuation of International Application No. PCT/KR2021/015908, filed on Nov. 4, 2021, which claims priority to Korean Patent Application No. 10-2020-0150360, filed on Nov. 11, 2020 and Korean Patent Application No. 10-2021-0001528, filed on Jan. 6, 2021 in the Korean Intellectual Property Office, the disclosures of which are herein incorporated by reference.

TECHNICAL FIELD
(1) Certain embodiments of the disclosure relate to an electronic device and a method for displaying a user interface of an electronic device.
BACKGROUND
(2) With the development of digital technology, electronic devices have diversified into a variety of device types, including smartphones, laptops, tablets, etc. in addition to traditional personal computers. Modern electronic device design further tends towards portability and larger display screens, to enhance convenience for end users.
(3) Electronic devices may be implemented with expandable display areas. For example, a part of a flexible display may extend outwards from within an inner space of an electronic device, thereby increasing screen area.
(4) Meanwhile, an electronic device may include an "angle of arrival" or AOA-based antenna in order to identify a position relative to an external electronic device.
SUMMARY

(5) An electronic device may include an AOA-based antenna disposed thereon, so as to interwork with a lower portion of a flexible display. If the screen area of the flexible display is expandable via sliding, the directivity of the AOA-based antenna may change disadvantageously.
(6) Certain embodiments of the disclosure provide an electronic device and a method for displaying a user interface of an electronic device, such that when directivity of the AOA-based antenna changes according to the sliding operation, and a sharing operation is executing between the electronic device and an external electronic device, a user interface related to the sharing operation may be displayed.
(7) An electronic device according to certain embodiments may include: a display including a first screen area that is maintained in exposure to an external environment, and an extended screen area that is exposed or stowed, according to a change in a state of the electronic device, communication circuitry configured to transceive a signal to or using an angle-of-arrival (AOA) antenna disposed within the extended screen area, a memory, a processor operatively coupled to the display, the AOA antenna, the communication circuitry, and the memory, wherein the memory stores instructions executable by the processor to cause the electronic device to: determine a position of the electronic device relative to an external electronic device, using the communication circuitry, execute a sharing operation with the external electronic device, identify whether the display of the electronic device is disposed in a first state or a second state, based on detecting that the display is disposed in the first state, perform a first positioning operation for the external electronic device, and display a sharing interface according to the first positioning operation on the display.
(8) A method according to certain embodiments may include: executing a sharing operation to transceive data with an external electronic device using communication circuitry including an angle-of-arrival (AOA) antenna disposed, detecting whether a display of the electronic device is disposed in a first state or second state according to whether an extendable screen area of the display is stowed within the electronic device, or extended to be exposed to an external environment of the electronic device, executing, by a processor, a first positioning operation for the external electronic device based on detecting the display is in the first state, and displaying a sharing interface according to the first positioning operation on the display.
(9) The electronic device and method (and variations thereof) disclosed herein are advantageous in that, when the directivity of an AOA-based antenna is changed by extension and/or retraction of a display via the sliding operation, a user interface related to sharing between the electronic device and an external electronic device may be displayed.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a block diagram illustrating an electronic device in a network environment.
(2) FIG. **2** is a flowchart illustrating an example method of displaying a user interface of an electronic device according to certain embodiments of the disclosure.
(3) FIG. **3** is a flowchart illustrating an example method of displaying a user interface of an electronic device according to certain embodiments of the disclosure.
(4) FIG. **4** is a flowchart illustrating an example method of displaying a user interface of an electronic device according to certain embodiments of the disclosure.
(5) FIG. **5**A is a diagram illustrating an example user interface of an electronic device in a first state according to certain embodiments of the disclosure.
(6) FIG. **5**B is a diagram illustrating an example user interface of an electronic device in a second state according to certain embodiments of the disclosure.
(7) FIG. **6** is a diagram illustrating an example user interface when a predetermined time has elapsed in FIG. **5**B.

(8) FIGS. **7**A to **7**C are diagrams illustrating an example state change of an electronic device according to certain embodiments of the disclosure.
(9) FIGS. **8**A and **8**B are diagrams illustrating an example method of detecting a state change of an electronic device according to certain embodiments of the disclosure.
(10) FIG. **9** is a diagram illustrating an example method of detecting a state change of an electronic device according to certain embodiments.
(11) FIG. **10** is a diagram illustrating an example curvature of an AOA antenna according to certain embodiments of the disclosure.
(12) FIG. **11** is a diagram illustrating an example layer of a display module according to certain embodiments of the disclosure.
(13) FIG. **12** is a diagram illustrating an example AOA antenna according to certain embodiments of the disclosure.
(14) FIG. **13** is a diagram illustrating an example configuration and operation of an AOA antenna according to certain embodiments of the disclosure.
(15) FIG. **14** is a diagram illustrating an example configuration and operation of an AOA antenna according to certain embodiments of the disclosure.
(16) FIG. **15** is a diagram illustrating an example operation of an AOA antenna of an electronic device according to certain embodiments of the disclosure.
(17) FIG. **16** is a diagram illustrating an example aspect ratio of a display module and the arrangement of an AOA antenna according to the disclosure.
(18) FIG. **17** is a diagram illustrating an example aspect ratio of a display module and the arrangement of an AOA antenna according to the disclosure.
(19) FIG. **18** is a block diagram illustrating an example method of displaying a user interface of an electronic device according to certain embodiments of the disclosure.
(20) FIG. **19** is a diagram illustrating an example electronic device **101** including a method of displaying the user interface of FIG. **18**.
(21) FIG. **20** is a diagram illustrating an example configuration of an electronic device according to certain embodiments of the disclosure.
(22) FIG. **21** is a block diagram illustrating an example electronic device according to certain embodiments of the disclosure.

DETAILED DESCRIPTION

(23) FIG. **1** is a block diagram illustrating an electronic device **101** in a network environment **100** according to certain embodiments. Referring to FIG. **1**, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).
(24) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation.

According to an embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.
(25) The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.
(26) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thererto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.
(27) The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.
(28) The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).
(29) The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.
(30) The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram

device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.
(31) The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.
(32) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.
(33) The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.
(34) A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).
(35) The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.
(36) The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.
(37) The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).
(38) The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.
(39) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external

electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

(40) The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(41) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element implemented using a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

(42) According to certain embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(43) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(44) According to an embodiment, commands or data may be transmitted or received between the

electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

(45) The electronic device according to certain embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

(46) It should be appreciated that certain embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

(47) As used in connection with certain embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(48) Certain embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory

136 or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a complier or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. The term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.
(49) According to an embodiment, a method according to certain embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.
(50) According to certain embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to certain embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to certain embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to certain embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.
(51) FIG. **2** is a flowchart illustrating a method of displaying a user interface of an electronic device **101** according to certain embodiments of the disclosure.
(52) In operation **201**, the electronic device **101** may perform a sharing operation with an external electronic device under the control of the processor **120**. The electronic device **101** may activate an AOA (angle of arrival) antenna included in the electronic device **101** for sharing with the external electronic device. In certain embodiments, the electronic device **101** may discover and/or measure the location of the external electronic device using the AOA antenna included in the electronic device **101** to share data with the external electronic device. For example, the electronic device **101** may share an image, a screen, and/or a communication with an external electronic device. The electronic device **101** may execute and display a user interface related to a sharing operation in order to share an image, screen, and/or communication with the external electronic device.
(53) The electronic device **101** may identify the state of the electronic device **101** in operation **203** under the control of the processor **120**.
(54) The electronic device **101** may extend or retract the display module **160** including the flexible display.
(55) When the display module **160** including the flexible display performs a rolling operation to reduce the display module **160**, the electronic device **101** may be in at least one of an open state, an extended state, and a rolling state.

(56) The rolling operation of the display module **160** may include an operation in which the electronic device **101** retracts the display module **160** into the housing thereof for stowage.
(57) When the display module **160** is extended by execution the sliding operation on the display module **160** including the flexible display, the electronic device **101** may be in at least one of a closed state, a reduced state, and a sliding state.
(58) The sliding operation of the display module **160** may include an operation in which the electronic device **101** extends the display module **160** as to exit at least in part from an interior of the housing thereof, becoming visible and/or accessible to an external environment of the device.
(59) In operation **205**, the electronic device **101** may display a positioning operation and a sharing interface on the display module **160** according to the identified state under the control of the processor **120**.
(60) In certain embodiments, the electronic device **101** may perform a positioning operation using the AOA antenna according to the state of the electronic device **101**.
(61) In certain embodiments, the electronic device **101** may perform a ranging operation using the AOA antenna according to the state of the electronic device **101**.
(62) In certain embodiments, the electronic device **101** may perform an AOA measurement operation including distance measurement and direction measurement by using an AOA antenna according to the state of the electronic device **101**.
(63) For example, the electronic device **101** may measure the relative position and/or direction of the external electronic device spaced apart from the electronic device **101** by using the AOA antenna.
(64) The directional position of the AOA antenna may be changed according to the state of the electronic device **101**.
(65) The AOA antenna may be a directional antenna using a patch antenna rather than an omni-directional antenna. The AOA antenna may be oriented in a direction perpendicular to the surface of the display module **160**.
(66) For example, when the electronic device **101** is in a closed state, the AOA antenna may face the rear surface of the electronic device **101**. When the electronic device **101** is in the closed state, the electronic device **101** may measure the relative position and/or direction of the external electronic device in the rear direction of the electronic device **101**. For example, when the electronic device **101** is in an open state, the AOA antenna may face the front surface of the electronic device **101**. When the electronic device **101** is in the open state, the electronic device **101** may measure the relative position and/or direction of the external electronic device in the front direction of the electronic device **101**.
(67) In certain embodiments, the AOA antenna may face the lateral direction of the electronic device **101** according to movement (or rolling) of the display module **160**. When the AOA antenna is disposed on the side surface of the electronic device **101**, the electronic device **101** may measure the relative position and/or direction of the external electronic device in the lateral direction of the electronic device **101**. In certain embodiments, the electronic device **101** may display a sharing interface on the display module **160** according to the state of the electronic device **101**.
(68) In certain embodiments, the AOA antenna may face the rear surface of the electronic device **101** when the electronic device **101** is in the closed state. When the electronic device **101** is in the closed state, the electronic device **101** may measure the relative position and/or direction of the external electronic device in the rear direction of the electronic device **101** and may display information on the relative position and/or direction of the external electronic device in the rear direction of the electronic device **101** on the display module **160** through a user interface.
(69) In certain embodiments, the AOA antenna may face the front surface of the electronic device **101** when the electronic device **101** is in the open state. When the electronic device **101** is in the open state, the electronic device **101** may measure the relative position and/or direction of the external electronic device in the front direction of the electronic device **101** and may display

information on the relative position and/or direction of the external electronic device in the front direction of the electronic device **101** on the display module **160** through the user interface.
(70) In operation **207**, the electronic device **101** may determine whether the state of the electronic device **101** is changed under the control of the processor **120**.
(71) When the state of the electronic device **101** is changed under the control of the processor **120**, the electronic device **101** may proceed from operation **207** to operation **209**.
(72) When the state of the electronic device **101** remains under the control of the processor **120** without being changed, the electronic device **101** may proceed from operation **207** to operation **205**.
(73) In certain embodiments, the state of the electronic device **101** may be changed to a second state (e.g., open state) by a user's manipulation or automatically while the electronic device **101** is displaying the sharing interface on the display module **160** in a first state (e.g., closed state).
(74) In operation **209**, when the state of the electronic device **101** is changed, the electronic device **101** may display the positioning operation and the sharing interface on the display module **160** according to the changed state under the control of the processor **120**.
(75) In certain embodiments, when the electronic device **101** is changed from the open state to the closed state, the electronic device **101** may measure the relative position and/or direction of the external electronic device in the rear direction of the electronic device **101** and may display information on the relative position and/or direction of the external electronic device in the rear direction of the electronic device **101** on the display module **160** through the user interface. In this case, the electronic device **101** may display the positioning operation and the user interface on the display module **160** for a predetermined time before the state change.
(76) For example, when the electronic device **101** is changed from the first state to the second state, a second user interface related to the second state may be displayed on the display module **160**. In this case, the electronic device **101** may display, on the display module **160**, the positioning operation and a first user interface in the first state together with the second user interface in the second state for a predetermined time.
(77) When the electronic device **101** is changed from the second state to the first state, the first user interface related to the first state may be displayed, on the display module **160**, the positioning operation and the second user interface in the second state together with the first user interface in the first state for a predetermined time.
(78) For example, when the electronic device **101** is changed from the open state to the closed state, the user interface related to the closed state may be displayed on the display module **160**. In this case, the electronic device **101** may display the positioning operation and the user interface in the open state on the display module **160** together with the user interface in the closed state for a predetermined time.
(79) In certain embodiments, when the electronic device **101** is changed from the closed state to the open state, the electronic device **101** may measure the relative position and/or direction of the external electronic device in the front direction of the electronic device **101**, and may display information on the relative position and/or direction of the external electronic device in the front direction of the electronic device **101** on the display module **160** through the user interface. For example, when the electronic device **101** is changed from the closed state to the open state, the electronic device **101** may display the user interface related to the open state on the display module **160**. In this case, the electronic device **101** may display the positioning operation and the user interface in the closed state together with the user interface in the open state for a predetermined time on the display module **160**.
(80) FIG. **3** is a flowchart illustrating a method of displaying the user interface of the electronic device **101** according to certain embodiments of the disclosure.
(81) In operation **301**, the electronic device **101** may perform a sharing operation with an external electronic device under the control of the processor **120**. An AOA antenna included in the

electronic device **101** may transmit/receive a positioning request message (e.g., a poll message or ranging message) and a positioning response message (e.g., a response message or ranging response message). The AOA antenna may include at least two or more patch antennas disposed on the display module **160**. In certain embodiments, the AOA antenna may further include an antenna (e.g., frame antenna, FPCB antenna, or LDS antenna) included in the electronic device **101** as well as at least two or more patch antennas disposed on the display module **160**.
(82) Since the antenna (e.g., frame antenna, FPCB antenna, or LDS antenna) included in the electronic device **101** has a positioning effect of the patch antenna due to a direction of the display module **160**, the antenna may be an omni-directional antenna for compensate for this.
(83) In operation **303**, the electronic device **101** may determine whether to be in a first state under the control of the processor **120**. The first state of the electronic device **101** may be, for example, a closed state. The second state of the electronic device **101** may be, for example, an open state.
(84) When the electronic device **101** is in the first state under the control of the processor **120**, the electronic device **101** may proceed from operation **303** to operation **305**.
(85) When the electronic device **101** is in the second state under the control of the processor **120**, the electronic device **101** may proceed from operation **303** to operation **309**.
(86) In operation **305**, the electronic device **101** may display a first positioning operation corresponding to the first state of the electronic device **101** and a first sharing interface on the display module **160** under the control of the processor **120**.
(87) The AOA antenna may face the rear surface of the electronic device **101** when the electronic device **101** is in the first state. When the electronic device **101** is in the first state, the electronic device **101** may measure the relative position and/or direction of the external electronic device in the rear direction of the electronic device **101** through the first positioning operation. Information on the relative position and/or direction of the external electronic device in the rear direction of the electronic device **101** may be displayed on the display module **160** through the first sharing interface.
(88) In operation **307**, the electronic device **101** may determine whether the electronic device **101** is changed from the first state to the second state under the control of the processor **120**.
(89) When the state of the electronic device **101** is changed to the second state under the control of the processor **120**, the electronic device **101** may proceed from operation **307** to operation **309**.
(90) When the state of the electronic device **101** remains under the control of the processor **120** without being changed, the electronic device **101** may proceed from operation **307** to operation **305**.
(91) In certain embodiments, the state of the electronic device **101** may be changed to the second state through a user's manipulation and/or automatically while the electronic device **101** is displaying the sharing interface on the display module **160** in the first state.
(92) When the electronic device **101** is changed to the second state, in operation **309**, the electronic device **101** may display a second positioning operation corresponding to the second state and a second sharing interface on the display module **160** under the control of the processor **120**.
(93) The AOA antenna may face the front surface of the electronic device **101** when the electronic device **101** is in the second state. When the electronic device **101** is in the second state, the electronic device **101** may measure the relative position and/or direction of the external electronic device in the front direction of the electronic device **101** through the second positioning operation. Information on the relative position and/or direction of the external electronic device in the front direction of the electronic device **101** may be displayed on the display module **160** through the second sharing interface.
(94) In certain embodiments, when the electronic device **101** is changed from the first state to the second state, the electronic device **101** may display the relative position and/or direction of the external electronic device in the front direction of the electronic device **101**, and may display information on the relative position and/or direction of the external electronic device in the front

direction of the electronic device **101** on the display module **160** through the user interface. For example, when the electronic device **101** is changed from the first state to the second state, the electronic device **101** may display a second sharing interface on the display module **160**. In this case, the electronic device **101** may display information on the relative position and/or direction of the external electronic device in the rear direction obtained by the positioning operation in the first state, on the second sharing interface for a predetermined time.

(95) FIG. **4** is a flowchart illustrating a method of displaying the user interface of the electronic device **101** according to certain embodiments of the disclosure.

(96) In operation **401**, the electronic device **101** may perform a sharing operation with an external electronic device under the control of the processor **120**.

(97) In operation **403**, the electronic device **101** may determine whether the electronic device **101** is in a second state under the control of the processor **120**.

(98) When the electronic device **101** is in the second state under the control of the processor **120**, the electronic device **101** may proceed from operation **403** to operation **405**.

(99) When the electronic device **101** is in a first state under the control of the processor **120**, the electronic device **101** may proceed from operation **403** to operation **409**.

(100) In operation **405**, the electronic device **101** may display a second positioning operation corresponding to the second state of the electronic device **101** and a second sharing interface on the display module **160** under the control of the processor **120**.

(101) In operation **407**, the electronic device **101** may determine whether the electronic device **101** is changed from the second state to the first state under the control of the processor **120**.

(102) When the state of the electronic device **101** is changed to the first state under the control of the processor **120**, the electronic device **101** may proceed from operation **407** to operation **409**.

(103) When the state of the electronic device **101** remains under the control of the processor **120** without being changed, the electronic device **101** may proceed from operation **407** to operation **405**.

(104) In certain embodiments, the state of the electronic device **101** may be changed to the first state by a user's manipulation or automatically while the electronic device **101** is displaying the sharing interface on the display module **160** in the second state.

(105) When the electronic device **101** is changed to the first state, in operation **409**, the electronic device **101** may display a first positioning operation corresponding to the first state and a first sharing interface on the display module **160** under the control of the processor **120**.

(106) An AOA antenna may face the rear surface of the electronic device **101** when the electronic device **101** is in the first state. When the electronic device **101** is in the second state, the electronic device **101** may measure the relative position and/or direction of the external electronic device in the front direction of the electronic device **101** through a second positioning operation. Information on the relative position and/or direction of the external electronic device in the front direction of the electronic device **101** may be displayed on the display module **160** through the second sharing interface.

(107) In certain embodiments, when the electronic device **101** is changed from the second state to the first state, the electronic device **101** may display the relative position and/or direction of the external electronic device in the rear direction of the electronic device **101**, and may display information on the relative position and/or direction of the external electronic device in the rear direction of the electronic device **101** on the display module **160** through the user interface. For example, when the electronic device **101** changes from the second state to the first state, the electronic device **101** may display the first sharing interface on the display module **160**. In this case, the electronic device **101** may display the information on the relative position and/or direction of the external electronic device in the front direction obtained through the positioning operation in the second state on the first sharing interface for a predetermined time.

(108) FIG. **5**A is a diagram illustrating the user interface of the electronic device **101** in a first state

according to certain embodiments of the disclosure.
(109) FIG. 5B is a diagram illustrating the user interface of the electronic device **101** in a second state according to certain embodiments of the disclosure.
(110) In FIG. 5A, the electronic device **101** is disposed in a first state (or a closed state). The electronic device **101** is slidable and can extend the display module **160** is in the first state. Initially here, the electronic device **101** may display a user interface through the display module **160** having a basic screen **501** (e.g., a first or initial screen area). The basic screen **501** may be expandable by sliding, and may include a minimum screen of the electronic device **101** prior to extending the display module **160**.
(111) In FIG. 5B, the electronic device **101** is disposed in a second state (or an open state). When the electronic device **101** slides relative to itself, as to extend the display module **160** into the second state, the electronic device **101** may then display the user interface through the display module **160** on both the basic screen area **501** and an extended screen area **502**.
(112) In FIG. 5B, when the electronic device **101** enters the second state (or the open state), an AOA antenna **730** disposed below the display module **160** may be directed from the rear direction of the electronic device **101** to the front direction thereof.
(113) The user interface may include an interface **510** (e.g., a first display region) that is related to the direction of the AOA antenna, and/or an interface **520** (e.g., a second display region) related to an external electronic device with which information sharing is to be executed.
(114) The interface **510** for the AOA antenna direction may include an AOA antenna indicator **511** and an interface **512** for a first corresponding device (e.g., icons **511**, **512**). The AOA antenna indicator **511** may indicate a direction in which the AOA antenna is directed with respect to the electronic device **101**. The interface **512** for the first corresponding device may include an interface (or an icon) indicating the electronic device **101**, and/or the electronic device **101** within a predetermined distance with respect to the AOA antenna, and/or a device correspond to (matching) the AOA.
(115) In FIGS. 5A and 5B, an external electronic device “K” (the first corresponding device) may be located within a minimum directivity angle (about +/−5 degrees) with respect to the AOA antenna of the electronic device **101**, and may thus be positioned on a directional line directed by the electronic device **101**. The AOA antennas may function with detection reliability for +/−60 degree measurements. In FIG. 5A, the electronic device **101** may select the external electronic device “K” for recommendation corresponding to a “frontal” direction of the AOA antenna and/or the direction directed by the electronic device **101**, and may display the recommended external electronic device “K” using the interface **512** representative of the first corresponding device, thereby facilitating convenient selection the first corresponding device for sharing data.
(116) In FIG. 5A, the external electronic device “K” is selected for recommendation because the external electronic device “K”, which is the first corresponding device, is disposed within the minimum directional angle of the AOA antenna. However, if the directional angle of the electronic device **101** is rotated to left/right, a tablet “J” and a laptop “K” may be recommended instead as they transition into the minimum directional angle of the AOA antenna.
(117) The electronic device **101** may display slide control icons **541** and **542** (e.g., “sliding controllers”), which may be related to sliding-in and/or sliding-out on the display module **160**. The sliding controllers **541** and **542** may indicate a slid-in and/or slid-out state of the display module **160** of the electronic device **101**. In certain embodiments, the sliding controllers **541** and **542** may be selectable by a user input, and in response to selection, may trigger actuation of the slide-in and/or slide-out state of the display module **160** for the electronic device **101**.
(118) In FIG. 5A, the electronic device **101** may display the interface **510** related to the AOA antenna direction, and the interface **512** related to the first corresponding device on the display module **160**. In this case, when the electronic device **101** is switched to the second state (whether manually by the user, or automatically by mechanism), an interface **513** related to a second

corresponding device may be displayed as shown in FIG. **5**B. In certain embodiments, in FIG. **5**B, the electronic device **101** may simultaneously display the interface **512** related to the first corresponding device, and the interface **513** related to the second corresponding device on the display module **160** for a predetermined time.
(119) In certain embodiments, the electronic device **101** may display a timer **550** related to a predetermined time on the interface **510** related to the AOA antenna direction.
(120) The interface **520** for a shareable external electronic device may include an interface **521** (e.g., a selectable icon) for a shareable external electronic device in a first state and an interface **522** (e.g., another selectable icon) for a shareable external electronic device in a second state. The interface **520** related to the shareable external electronic device may prioritize the interface **521** related to the shareable external electronic device in the first state and the interface **522** related to the shareable external electronic device in the second state based on the state of the electronic device **101**, and may display the prioritized interfaces **521** and **522** on the display module **160**.
(121) For example, as in FIG. **5**A, when the electronic device **101** is in the first state, the interface **520** related to the sharable external electronic device may arrange the interface **521** related to the sharable external electronic device in the first state in the first row and may arrange the interface **522** related to the sharable external electronic device in the second state in the second row.
(122) For example, as in FIG. **5**B, when the electronic device **101** is in the second state, the interface **520** related to external electronic devices with which sharing may be executed, may disposed the interface **522** (e.g., the icon) related to the sharable external electronic device “M” in the second state in the first row and may arrange the interface **521** related to the sharable external electronic device in the first state in the second row “K””.
(123) In FIG. **5**B, when the display module **160** of the electronic device **101** is extended and is in the second state, the interface **510** related to the AOA antenna direction may be extended, and the interface **513** related to the second corresponding device may be displayed while the size and position of the interface **520** related to the shareable external electronic device are maintained. In FIG. **5**A, at least one or more sharing device icons **530** included in the interface **520** related to the shareable external electronic device may further include a directional indicator **531** and text **532**. The sharing device icon **530** may display a character or image (e.g., an image configured by the user) representing a user's name of a shareable device on the icon. In addition, the sharing device icon **530** may at least partially include the direction indicator **531** indicating a relative direction (or azimuth) from the electronic device **101**. The text **532** may further display information indicating the user's name of the shareable device and/or the type of the device (e.g., laptop, tablet, or mobile phone).
(124) In FIG. **5**A, when the electronic device **101** is in the first state for a predetermined time or longer, the electronic device **101** may cause the interface **522** related to the sharable external electronic device in the second state to be removed from display on the display module **160**, and may maintain display of the interface **521** related to the sharable external electronic device in the first state to be displayed on the display module **160**.
(125) FIG. **6** is a diagram illustrating a user interface when a predetermined time has elapsed in FIG. **5**B.
(126) The electronic device **101** may display the user interface as shown in FIG. **5**B on the display module **160** after the device is switched from the first state to the second state. At this time, when a predetermined time (e.g., 30 seconds) has elapsed, an AOA antenna **730** may track the location of the external electronic device **513** in the second state, as this tracking directionality is more readily function in the second state than in the first state. When a predetermined time has elapsed after being switched to the second state, the electronic device **10** may terminate display of the interface **512** related to the first corresponding device, and may maintain display of the interface **510** related to the AOA antenna directionality, and the interface **513** related to a second corresponding device on the display module **160**.

(127) In certain embodiments, while continuously tracking the location of the device measured in the first state using an omni-directional antenna (e.g., frame antenna) disposed in a side frame of the electronic device **101**, the electronic device **101** may track the location of the external electronic device in the second state. In this case, even if a predetermined time (e.g., 30 seconds) has elapsed after the electronic device **101** is switched from the first state to the second state, as shown in FIG. 5B, the interface **512** for the first corresponding device and the interface **513** for the second corresponding device may be simultaneously displayed on the display module **160**.
(128) FIGS. **7**A to **7**C are diagrams illustrating the state change of the electronic device **101** according to certain embodiments of the disclosure.
(129) The display module **160** may include a display panel **161**, a shielding member **711**, and a metal plate **712**.
(130) The display panel **161** may display an image and/or a screen as a flexible display.
(131) The display panel **161** may face a front surface **701** of the electronic device **101**, and the metal plate **712** may face a rear surface **702** of the electronic device **101**.
(132) A shielding member **711** may serve as step difference compensation when the display panel **161** is coupled to the metal plate **712**. The shielding member **711** may be implemented using a conductive member that prevents the components of the electronic device **101** disposed below the metal plate **712** from affecting the display panel **161**.
(133) The metal plate **712** may be combined with a sliding structure **720**. The metal plate **712** may be moved when the sliding structure **720** is rotated. The metal plate **712** may be coupled to the display panel **161** to move together.
(134) In FIG. **7**A, when the electronic device **101** is in a first state (e.g., closed state), the AOA antenna **730** may be oriented to face the rear surface **702** of the electronic device **101**. The electronic device **101** may display the basic screen **501** in the first state.
(135) FIG. **7**B is a diagram illustrating the AOA antenna **730** in a bending section of the flexible display, when the electronic device **101** is extended in the first state (or reduced in the second state).
(136) In FIG. **7**B, when the electronic device **101** is extended in the first state (or reduced in the second state), the AOA antenna **730** may be disposed within the bending section. When the AOA antenna **730** is moved by the sliding action, bending stress may be reduced and the curvature length may be limited to prevent separation of the AOA antenna **730** from the display module **160**. Accordingly, the AOA antenna **730** may formed to have a length of curvature smaller than the curvature length of the display panel **161** or the display module **160** in the bending section.
(137) When the electronic device **101** is extended in the first state (or reduced in the second state), the electronic device **101** may further display a screen area, as extended by a first extended length d**1**.
(138) In FIG. **7**C, when the electronic device **101** is in the second state (e.g., open state), the AOA antenna **730** may be oriented to face the front surface **701** of the electronic device **101**. When the electronic device **101** transitions into the second state, the electronic device **101** may add and display the extended screen area **502** corresponding to a second extended length d**2**.
(139) FIGS. **8**A and **8**B are diagrams illustrating a method of detecting the state change of the electronic device **101** according to certain embodiments of the disclosure.
(140) FIGS. **8**A to **8**B have the same components as FIGS. **7**A to **7**B, and additionally show components such as a first magnet **821** and a second magnet **822**, a printed circuit board **810**, the processor **120**, and a hall effect sensor **820**.
(141) Referring to FIG. **8**A, the electronic device **101** may include a Hall sensor **820** disposed on the printed circuit board **810**. The Hall sensor **820** may transmit information regarding a detected magnetic force change to the processor **120**. The printed circuit board **810** may be disposed in a lower area of the display module **160** (or in the housing of the electronic device **101**). The first magnet **821** and the second magnet **822** may be disposed on at least a portion of the metal plate

712. The second magnet **822** may be disposed on at least a portion of the metal plate **712** corresponding to the Hall sensor **820**, and the first magnet **821** may be disposed on the at least a portion of the metal plate **712** spaced apart from the second magnetic **822** by the second extended length d**2**. In this case, as shown in FIG. **8**A, when the electronic device **101** is in the first state (closed state), the first magnet **821** does not align to the printed circuit board **810**, and instead aligns with the at least a portion of the metal plate **712** disposed on the rear surface **702**. The processor **120** disposed on the printed circuit board **810** may be disposed to be spaced apart from the Hall sensor **820** by the second extended length d**2**.

(142) Referring to FIG. **8**B, when the electronic device **101** is changed from the first state to the second state, the first magnet **821** may align with the Hall sensor **820**, and the second magnet **822** may align the processor **120**. When the electronic device **101** is changed from the first state to the second state, the display module **160** may move by the second extended length d**2**. Because the printed circuit board **810** fixed to the housing of the electronic device **101** does not move, the first magnet **821** and the second magnet **822** disposed on the metal plate **712** of the display module **160** may move relative to the Hall sensor **820** and the processor **120**.

(143) Thus, the processor **120** may detect the state of the electronic device **101** based on information on the open state and/or closed state, according to the magnetic force change detected by the Hall sensor **820** and/or the processor **120** via alignment with either the first magnet **821** or the second magnet **822**.

(144) FIG. **9** is a diagram illustrating a method of detecting the state change of the electronic device **101** according to certain embodiments.

(145) FIG. **9** may have the same components as FIGS. **7**A to **7**B, and also additionally include components such as a detection pattern **910**, a printed circuit board **810**, the processor **120**, a first optical sensor **911**, and a second optical sensor **912**. When a state change of the electronic device **101** are detectable using the detection pattern **910**, the electronic device **101** may recognize not only the open state and/or closed state of the electronic device **101**, but also an intermediate state thereof in the midst of a transition to the open state and/or closed state.

(146) Referring to FIG. **9**, the electronic device **101** may include a first optical sensor **911** and a second optical sensor **912** on the printed circuit board **810**. The metal plate **712** may include the detection pattern **910** in at least a portion thereof. The detection pattern **910** may be disposed on at least a portion of the metal plate **712** and oriented to face the printed circuit board **810** included in the electronic device **101**. The detection pattern **910** may include a pattern length of the second extended length d**2**. The detection pattern **910** may include, for example, an uneven pattern. The first optical sensor **911** and the second optical sensor **912** may include infrared sensors. The first optical sensor **911** and the second optical sensor **912** may include, for example, an emitter that outputs infrared rays, and a receiver that receives reflected infrared rays. In certain embodiments, the first optical sensor **911** may include the emitter and the second optical sensor **912** may include the receiver. The first optical sensor **911** and the second optical sensor **912** may irradiate light towards the detection pattern **910**, and may transmit data obtained by counting a change in the reflected light to the processor **120**. The processor **120** may detect the state of the electronic device **101** based on the data received from the first optical sensor **911** and the second optical sensor **912**.

(147) In certain embodiments, when the electronic device **101** executes a sliding or rolling operation of the display module **160** using a driving unit (or a rolling actuator), the electronic device **101** may determine a slide length to determine whether the electronic device **101** is disposed in the open state, the closed state, and/or the intermediate state, based on the number of rotations of the motor of the driving unit and the rotation time thereof.

(148) FIG. **10** is a diagram illustrating the curvature of an AOA antenna according to certain embodiments of the disclosure.

(149) The AOA antenna **730** may have a curvature of a length smaller than a curvature length of the display panel **161** or the display module **160** in a bending section. In the bending section, the

bending length of the AOA antenna **730** may be R**1** and the bending length of the display module **160** may be R**2**. R**1** may be smaller than R**2**. The AOA antenna **730** may include a curvature smaller than the curvature of the display module **160**, so that it is possible to reduce the repulsive force of separation due to the curvature when the AOA antenna **730** is moving within the bending section.

(150) FIG. **11** is a diagram illustrating a layer of the display module **160** according to certain embodiments of the disclosure.

(151) The display module **160** may include a basic screen **501** (e.g., an initial or first screen area) that is exposed and visible from an exterior of the electronic device **101** (e.g., constantly exposed) and an extended screen **502** that is selectable exposed according to whether the electronic device **101** is disposed in the open or closed state. The extended screen **502** may be exposed to the exterior of the electronic device **101** in an open state of the electronic device **101**.

(152) The extended screen **502** may include a bending area **1101** and a planar area **1102**. The bending area **1101** is maintained in a bent state in the closed state of the electronic device **101**, and is continuous with a basic screen area. The planar area **1102** is an area that is maintained e, and is continuous with the bending area **1101**.

(153) When the electronic device **101** is transitioned from the closed state to the open state, or from the open state to the closed state, the planar area **1102** may be at least partially bent by rotating around the sliding structure **720**.

(154) In the closed state of the electronic device **101**, at least a portion (e.g., the planar area **1102**) of the extended screen **502** may be oriented to face the rear surface **702** of the electronic device **101**, and in the open state, the at least the portion (e.g., the planar area **1102**) thereof may be oriented to face the front surface **701** of the electronic device **101**.

(155) The sliding structure **720** may include a roller, and may be further operatively coupled to a gear structure **1130** included in the at least a portion of the metal plate **712** to facilitate linear movement of the display module **160** through the rotation.

(156) The display module **160** may include a plurality of layers, which may be arranged in a prespecified order, such as the display panel **161**, the shielding member **711**, and then the metal plate **712**. The display panel **161** may be configured to display a screen and other assorted imagery according to the control of the processor **120** and/or a display driver IC **1110**.

(157) The display panel **161** may include a transparent layer in an external exposure direction, and may further include an opaque layer (e.g., an “EMBO” layer) in an internal direction of the electronic device **101**. The transparent layer may protect the surface of the display panel **161**, and the opaque layer may prevent a component (e.g., the AOA antenna **730**) disposed below the display panel **161** from being visible from an exterior of the electronic device.

(158) The display panel **161** may be coupled to the shielding member **711** using an adhesive member **1120**.

(159) The shielding member **711** may provide step difference compensation when the display panel **161** is coupled to the metal plate **712**. The shielding member **711** may be conductive, which may prevent components of the electronic device **101** disposed below the metal plate **712** from interfering with the display panel **161**.

(160) The shielding member **711** may include the display driver IC **1110** and the AOA antenna **730** in at least a partial area thereof. The display driver IC **1110** may be disposed on the basic area **501**. The AOA antenna **730** may be disposed on the extended screen **502**.

(161) The shielding member **711** may be coupled to the display panel **161** and the metal plate **712** using an adhesive member **1120**.

(162) The metal plate **712** may be formed a separate plate structure, and may include a multi-bar structure, and a general plate portion lacking a multi-bar structure, the two being mechanically combined with each other.

(163) In certain embodiments, the metal plate **712** may be implemented as a multi-hole structure on

a thin plate, and some holes may be rotationally moved in engagement with a roller (e.g., the sliding structure **720**).
(164) The metal plate **712** may be coupled to the shielding member **711** using the adhesive member **1120**.
(165) FIG. **12** is a diagram illustrating the AOA antenna **730** according to certain embodiments of the disclosure.
(166) The AOA antenna **730** may include a plurality of antenna elements disposed on a flexible printed circuit board (FPCB, **733**). The AOA antenna **730** may include a first antenna element **731** and a second antenna element **732**. In certain embodiments, the AOA antenna may further include a third antenna element.
(167) Each of the first antenna element **731** and the second antenna element **732** may be implemented using multiple layers. The first antenna element **731** and the second antenna element **732** may be patch antennas. The first antenna element **731** and the second antenna element **732** may include a ground layer **1211** and a signal layer **1212**. The ground layer **1211** may have a mesh shape. In certain embodiments, the ground layer **1211** may remove a conductive member on grid lines and may form the conductive member on the remaining portion. The signal layer **1212** may form a crosshair-shaped no-signal area and a butterfly-shaped signal area with square wings.
(168) FIG. **13** is a diagram illustrating the configuration and operation of the AOA antenna **730** according to certain embodiments of the disclosure.
(169) The display panel **161** may further include a transparent layer **163** and an opaque layer (e.g., an EMBO layer).
(170) The shielding member **711** may include the AOA antenna **730** in at least a partial area thereof. The AOA antenna **730** may include a plurality of antenna elements **731** and **732**. The AOA antenna **730** may include a directional antenna. Each of the first antenna element **731** and the second antenna element **732** may include a patch antenna. The first antenna element **731** may output a first beam A, and the second antenna element **732** may output a second beam B. A directivity direction and a directivity angle may be obtained within a range of −60 degrees to +60 degrees with respect to the first beam A and the second beam B. A field of view (FOV) of the AOA antenna **730** may correspond to the beam directivity angles for the first beam A and the second beam B.
(171) The shielding member **711** may be disposed between the display panel **161** and the metal plate **712**. In certain embodiments, the AOA antenna **730** may be disposed between the display panel **161** and the metal plate **712**.
(172) In certain embodiments, the first antenna element **731** and the second antenna element **732** may perform a communication transmission/reception function. In certain embodiments, at least one of the first antenna element **731** and the second antenna element **732** may perform a communication reception function, and the other thereof may perform a communication transmission/reception function.
(173) FIG. **14** is a diagram illustrating the configuration and operation of the AOA antenna **730** according to certain embodiments of the disclosure.
(174) The AOA antenna **730** of FIG. **14** has the same components as that of FIG. **13**, but the arrangement therein of the AOA antenna **730** may be different, as will be discussed below.
(175) The AOA antenna **730** may be disposed below the display panel **161**. The shielding member **711** and the metal plate **712** may include the AOA antenna **730** in at least a partial area thereof. The metal plate **712** may further define an opening in which the AOA antenna **730** may be disposed. Since the metal plate **712** of FIG. **14** further defines the opening (e.g., unlike the metal plate **712** of FIG. **13**) for receiving the AOA antenna **730**, the AOA antenna **730** of FIG. **14** may thicker than the AOA antenna **730** of FIG. **13**. In this case, the relatively thicker AOA antenna **730** of FIG. **14** may demonstrate an increase in antenna gain in comparison to the AOA antenna **730** of FIG. **13**.
(176) FIG. **15** is a diagram illustrating an operation of the AOA antenna **730** of the electronic

device **101** according to certain embodiments of the disclosure.
(177) The AOA antenna **730** may include the FPCB **733**, the first antenna element **731** and the second antenna element **732**. The first antenna element **731** and the second antenna element **732** may be disposed on the FPCB **733**.
(178) At least one of the first antenna element **731** and the second antenna element **732** may operate as a communication transmission/reception channel, and the other thereof may operate as a reception channel.
(179) Each of the first antenna element **731** and the second antenna element **732** may be connected to a communication circuitry **1510** (e.g., the communication module **190** of FIG. **1**). The communication circuitry **1510** may include ultra-wide band communication circuitry. The communication circuitry **1510** may transmit a message related to location measurement to an external electronic device **1501** through the first antenna element **731** and/or the second antenna element **732** using an asynchronous communication method. When receiving the location measurement message from the electronic device **101**, the external electronic device **1501** may transmit a response message to the electronic device **101**. The electronic device **101** may receive the response message S**1** of the external electronic device **1501** through the first antenna element **731** and/or the second antenna element **732**. The communication circuitry **1510** may perform communication using, for example, one of four channels of a 6-8 GHz band.
(180) The communication circuitry **1510** may transmit phase information of the signal received through the first antenna element **731** and/or the second antenna element **732** to the processor **120**. The processor **120** may detect a relative direction between the external electronic device **1501** and the electronic device **101** based on the received phase information of the signal.
(181) The communication circuitry **1510** may transmit time information of the signal received through the first antenna element **731** and/or the second antenna element **732** to the processor **120**. The processor **120** may detect a relative distance between the external electronic device **1501** and the electronic device **101** based on the received time information of the signal.
(182) FIG. **16** is a diagram illustrating the aspect ratio of the display module **160** and the arrangement of the AOA antenna **730** according to the disclosure.
(183) In a closed state, the electronic device **101** may utilize for display the basic screen area **501**, having a first ratio W**1**. For example, the first ratio W**1** (i.e., a horizontal to vertical ratio) may be 16:9. In an open state, the electronic device **101** may display a screen using the basic screen area **501** and the extended screen area **502**. The basic screen **501** and the extended screen **502** may have a second ratio W**2**. For example, the second ratio W**2** may be 21:9.
(184) In certain embodiments, the electronic device **101** may extend the display module **160** via a sliding articulation. When the display module **160** is extended by sliding, the extended screen **502** may be extended, which in some configurations may be up to twice the size of the basic screen.
(185) The AOA antenna **730** disposed in the extended screen (e.g., the extended screen area) and/or the planar area **1102** may, be disposed such that the first antenna element **731** and/or the second antenna element **732** are arranged in a line. The first antenna element **731** and/or the second antenna element **732** may be arranged in a line formation, oriented in a vertical and/or horizontal direction (e.g., row or column), to receive a signal. The processor **120** may utilize a phase difference between the signals received from the first antenna element **731** and/or the second antenna element **732** arranged in a line to obtain a relative direction (or azimuth) between the electronic device **101** and the external electronic device. The display module **160** may display a screen under the control of the display driver IC **1110**.
(186) FIG. **17** is a diagram illustrating the aspect ratio of the display module **160** and the arrangement of the AOA antenna **730** according to the disclosure.
(187) The AOA antenna **730** of FIG. **17** may further include a third antenna element **734** within the AOA antenna **730** of FIG. **16**, and the remaining components are the same.
(188) The AOA antenna **730** disposed in the extended screen **502** (e.g., the extended screen area)

and/or the planar area **1102** may include the first antenna element **731** and/or the second antenna element **732** arranged in a line. The AOA antenna **730** may include the second antenna element **732** and/or the third antenna element **734** arranged in a line. In certain embodiments, the first antenna element **731** and/or the second antenna element **732** may be arranged in a line in a first direction, and the second antenna element **732** and/or the third antenna element **734** may be arranged in a line in a second direction. The first direction and the second direction may vertically intersect on a plane. For example, the first direction may be a y-axis direction, and the second direction may be an x-axis direction.

(189) The first antenna element **731**, the second antenna element **732**, and/or the third antenna element **734** may be arranged in a line in a vertical and/or horizontal direction (e.g., as a column or a row) to receive signals. The processor **120** may use a phase difference detected between the signal received from the first antenna element **731**, the second antenna element **732**, and/or the third antenna element **734** (as arranged in a line) to obtain information on a relative direction (or azimuth) between the electronic device **101** and the external electronic device.

(190) When the electronic device **101** includes the first antenna element **731**, the second antenna element **732**, and/or the third antenna element **734** as the AOA antenna **730**, the patch antenna (the first antenna element **731**, the second antenna element **732**, and/or the third antenna element **734**) may be selectively activated in order to increase the positional accuracy of detection, for both the portrait mode and the landscape mode.

(191) For example, the electronic device **101** may activate a portion of the patch antenna (the first antenna element **731**, the second antenna element **732**, and/or the third antenna **734**) that aligns with an axis perpendicular to the vertical/horizontal mounting mode. When the electronic device **101** is disposed in the portrait mode, the first antenna element **731** and/or the second antenna element **732** may be activated. When the electronic device **101** is disposed in the landscape mode, the second antenna element **732** and/or the third antenna element **734** may be activated.

(192) In certain embodiments, the first antenna element **731**, the second antenna element **732**, and/or the third antenna element **734** may be arranged in a triangular shape to receive signals. The processor **120** may utilize a phase difference between the signals received from each of the first antenna element **731**, the second antenna element **732**, and/or the third antenna element **734** arranged in the triangular shape to obtain information on a relative direction (or azimuth) between the electronic device **101** and the external electronic device.

(193) In certain embodiments, the processor **120** may obtain information on the relative elevation between the electronic device **101** and the external electronic device, using a phase difference between the signals received from the first antenna element **731**, the second antenna element **732**, and/or the third antenna element **734** arranged in the line.

(194) In order to calculate the relative elevation between the electronic device **101** and the external electronic device, the first antenna element **731**, the second antenna element **732**, and/or the third antenna element **734** may simultaneously receive signals from the external electronic device.

(195) The electronic device **101** may obtain the information on the relative direction (azimuth) between the electronic device **101** and the external electronic device from signals received via two antennas (e.g., the first antenna element **731** and the second antenna element **732**) aligned in the vertical axis, and may obtain information on a relative elevation from the two antennas (e.g., the second antenna element **732** and/or the third antenna element **734**) as aligned in the horizontal axis. A method of obtaining AOA information using the first antenna element **731**, the second antenna element **732**, and/or the third antenna element **734** may be referred to as 3D AOA, and a method of obtaining AOA information using the first antenna element **731** and/or the second antenna element **732** may be referred to as “2D AOA.”

(196) FIG. **18** is a block diagram illustrating a method of displaying a user interface of the electronic device **101** according to certain embodiments of the disclosure.

(197) In operation **1801**, the electronic device **101** may perform a sharing operation with an

external electronic device under the control of the processor **120**. The electronic device **101** may execute an AOA (angle of arrival) antenna included in the electronic device **101** for sharing with the external electronic device. For example, the electronic device **101** may share an image, a screen, and/or communication with the external electronic device. The electronic device **101** may execute a user interface related to the sharing operation in order to share an image, screen, and/or communication with the external electronic device.

(198) In operation **1803**, the electronic device **101** may identify the state of the electronic device **101** under the control of the processor **120**. The electronic device **101** may extend or reduce the display module **160** including the flexible display. When the display module **160** of the electronic device **101** is extended, the electronic device **101** may be in an open state (or an extended state). When the display module **160** of the electronic device **101** is not extended and is in a basic state, the electronic device **101** may be in a closed state (or a basic state). The electronic device **101** may determine the extended and/or basic state of the display module **150** based on signals detected from sensors under the control of the processor **120**, and may identify the state of the electronic device **101**.

(199) In operation **1805**, the electronic device **101** may display a positioning operation and a sharing interface on the display module **160** according to the identified state under the control of the processor **120**.

(200) In certain embodiments, the electronic device **101** may perform a positioning operation using the AOA antenna according to the state of the electronic device **101**. For example, the electronic device **101** may measure the relative position and/or direction of the external electronic device spaced apart from the electronic device **101** using the AOA antenna.

(201) The directional position of the AOA antenna may be changed according to the state of the electronic device **101**. For example, when the AOA antenna is in the closed state of the electronic device **101**, the AOA antenna may face the rear surface of the electronic device **101**. When the electronic device **101** is in the closed state, the electronic device **101** may measure the relative position and/or direction of the external electronic device in the rear direction of the electronic device **101**.

(202) For example, when the AOA antenna is in the open state of the electronic device **101**, the AOA antenna may face the front surface of the electronic device **101**. When the electronic device **101** is in the open state, the electronic device **101** may measure the relative position and/or direction of the external electronic device in the front direction of the electronic device **101**.

(203) In certain embodiments, the electronic device **101** may display the sharing interface on the display module **160** according to the state of the electronic device **101**.

(204) In certain embodiments, the AOA antenna may face the rear surface of the electronic device **101** when the electronic device **101** is in the closed state. When the electronic device **101** is in the closed state, the electronic device **101** may measure the relative position and/or direction of the external electronic device in the rear direction of the electronic device **101** and may display information on the relative position and/or direction of the external electronic device on the display module **160** through the user interface.

(205) In certain embodiments, the AOA antenna may face the front surface of the electronic device **101** when the electronic device **101** is in the open state. When the electronic device **101** is in the open state, the electronic device **101** may measure the relative position and/or direction of the external electronic device in the front direction of the electronic device **101** and may display information on the relative position and/or direction of the external electronic device on the display module **160** through the user interface.

(206) In operation **1807**, the electronic device **101** may determine whether reception of a communication message from the external electronic device is detected under the control of the processor **120**.

(207) The communication message may be configured to establish communication linkage, and

may include, for example, a Bluetooth low energy (BLE) and/or Bluetooth communication pairing message. The BLE and/or Bluetooth communication pairing message may include an advertising packet.
(208) The advertising packet may include ultra-wideband (UWB) communication compatibility information of the external electronic device. The electronic device **101** may identify the UWB communication compatibility information included in the received advertising packet, may activate the AOA antenna **730**, and may perform a positioning operation using the AOA antenna **730**.
(209) When receiving a communication message from the external electronic device under the control of the processor **120**, the electronic device **101** may proceed from operation **1807** to operation **1809**.
(210) When the electronic device **101** does not receive the communication message from the external electronic device under the control of the processor **120**, the electronic device **101** may return from operation **1807** to operation **1805**.
(211) When receiving the communication message from the external electronic device, in operation **1809**, the electronic device **101** may determine whether an external electronic device corresponding to the communication message is detected through the AOA antenna under the control of the processor **120**.
(212) When the external electronic device corresponding to the communication message is detected through the AOA antenna under the control of the processor **120**, the electronic device **101** may proceed from operation **1809** to operation **1805**.
(213) When the external electronic device corresponding to the communication message is not detected through the AOA antenna under the control of the processor **120**, the electronic device **101** may proceed from operation **1809** to operation **1811**.
(214) In operation **1811**, the electronic device **101** may change the state of the electronic device **101** by actuating the driving unit under the control of the processor **120** to slide the display.
(215) The driving unit may include a mechanical device that changes the state of the electronic device **101** (e.g., by extending or retracting the display), and/or may rotate the sliding structure **720** of the electronic device **101** to change between the open state and/or the closed state.
(216) In operation **1811**, under the control of the processor **120**, the electronic device **101** may change the state of the electronic device **101** thereof from the closed state to the open state, or from the closed state to the closed state, via operation of the driving unit.
(217) In certain embodiments, when the state of the electronic device **101** changes from the closed state to the open state, the AOA antenna **730** facing the rear surface **702** may be directed toward the front surface **701** of the electronic device **101**.
(218) In certain embodiments, when the state of the electronic device **101** changes from the open state to the closed state, the AOA antenna **730** facing the front surface **701** may be directed toward the rear surface **702** of the electronic device **101**.
(219) In certain embodiments, when receiving the communication message from the external electronic device, in operation **1811**, the electronic device **101** may request the state change of the electronic device **101** from the user under the control of the processor **120** to change the state of the electronic device **101**.
(220) In certain embodiments, the state of the electronic device **101** may be changed by the user's manipulation and/or automatically while the electronic device **101** is displaying a sharing interface in the first state (e.g., closed state).
(221) In operation **1813**, the electronic device **101** may display the positioning operation and the sharing interface on the display module **160** according to the changed state under the control of the processor **120**.
(222) In certain embodiments, when the electronic device **101** is changed from the open state to the closed state, the electronic device **101** may measure the relative position and/or direction of the external electronic device in the rear direction of the electronic device **101**, and may display

information on the relative position and/or direction of the external electronic device in the rear direction of the electronic device **101** through the user interface. In this case, the electronic device **101** may display the positioning operation and the user interface on the display module **160** for a predetermined time before the state change.
(223) For example, when the electronic device **101** is changed from the open state to the closed state, a user interface related to the closed state may be displayed. In this case, the electronic device **101** may display the positioning operation and the user interface in the open state on the display module **160** together with the user interface in the closed state for a predetermined time.
(224) In certain embodiments, when the electronic device **101** is changed from the closed state to the open state, the electronic device **101** may measure the relative position and/or direction of the external electronic device in the front direction of the electronic device **101**, and may display information on the relative position and/or direction of the external electronic device in the front direction of the electronic device **101** through the user interface. For example, when the electronic device **101** is changed from the closed state to the open state, the electronic device **101** may display a user interface related to the open state. In this case, the electronic device **101** may display the positioning operation and the user interface in the closed state together with the user interface in the open state on the display module **160** for a predetermined time.
(225) FIG. **19** is a diagram illustrating the electronic device **101** including a method of displaying the user interface of FIG. **18**.
(226) When receiving a communication message from the external electronic device **1501**, the electronic device **101** may determine whether the external electronic device that transmitted the communication message is detected through the AOA antenna **730** under the control of the processor **120**.
(227) When the external electronic device **1501** corresponding to the communication message is detected through the AOA antenna **730**, the electronic device **101** may display a sharing interface corresponding to the current state (e.g., the open state or the closed state) of the electronic device **101** on the display module **160**.
(228) When the external electronic device **1501** corresponding to the communication message is not detected through the AOA antenna **730**, the electronic device **101** may change the state of the electronic device **101** using the driving units **1911** and **1912**.
(229) The electronic device **101** may use the driving units **1911** and **1912** to switch the state thereof from the closed state to the open state or from the open state to the closed state.
(230) In certain embodiments, when the state of the electronic device **101** is switched from the closed state to the open state, the AOA antenna **730** facing the rear surface **702** may be oriented to face the front surface **701** of the electronic device **101**. In certain embodiments, when the state of the electronic device **101** is switched from the open state to the closed state, the AOA antenna **730** facing the front surface **701** may be oriented face the rear surface **702** of the electronic device **101**.
(231) In certain embodiments, when receiving the communication message from the external electronic device, the electronic device **101** may request the state change of the electronic device **101** from the user to change the state of the electronic device **101**.
(232) In certain embodiments, whenever the state of the electronic device **101** is changed (e.g., when a sliding operation occurs), the electronic device **101** may perform a positioning operation through the AOA antenna.
(233) In certain embodiments, even when a new device is discovered or connected through BLE, the state of the electronic device **101** may be changed (e.g., the sliding operation occurs), and an automatic positioning operation may be performed through the AOA antenna.
(234) In certain embodiments, the electronic device **101** may include a first extended screen **1901** and/a second extended screen **1902**. In the open state, at least one of the first extended screen **1901** and/or the second extended screen **1902** may additionally appear, or both the first extended screen **1901** and/or the second extended screen **1902** may appear.

(235) The electronic device **101** may display the positioning operation and sharing interface **1920** on the display module **160** according to the changed state.
(236) FIG. **20** is a diagram illustrating the configuration of the electronic device **101** according to certain embodiments of the disclosure.
(237) The electronic device **101** may include a first housing **2010** and a second housing **2040**. The first housing **2010** is a main housing, and the second housing **2040** may include a driving unit (or a rolling actuator, **2021**). The first housing **2010** may include the sliding structure **720**, and a central axis of the sliding structure may be fixed to the first housing **2010**. When the electronic device **101** is switched to an open state or a closed state, the second housing **2040** may linearly move using the driving unit **2021** to cause the display module **160** to slide-in or slide-out. The driving unit **2021** may be disposed inside a space formed by the display module **160**, the first housing **2010**, and the second housing **2040**, and separation of the driving unit **2021** may be prevented by a support member **2020** included in the space. A cover **1030** (e.g., a glass cover) may be disposed on the rear surface **703** of the first housing **2010** and the second housing **2040**. The AOA antenna **730** may be oriented to face the front surface **701** when the electronic device **101** is in the open state, and may be oriented to face the rear surface **703** when the electronic device **101** is in the closed state.
(238) In certain embodiments, when the electronic device **101** performs a sliding or rolling operation of the display module **160** using the driving unit (or a rolling actuator, **2021**), an actual slide length of the display module **160** may be determined based on the number of rotations of the motor of the driving unit and the rotation time thereof, so that the open state, the closed state, and/or the intermediate state of the electronic device **101** may be identifiable to the processor.
(239) FIG. **21** is a block diagram illustrating the electronic device **101** according to certain embodiments of the disclosure.
(240) The electronic device **101** may include the processor **120**, the display module **160**, a positioning circuitry **2110**, a sensor circuitry **2120**, a near field communication circuitry **2130**, the first antenna **730**, a second antenna **2140**, and one or more driving units **1911** and **1912**.
(241) The display module **160** may slide-in or slide-out by the rotational motion of the one or more driving units **1911** and **1912** driven under the control of the processor **120**. The display module **160** may include the first antenna **730** in the extended area. The first antenna **730** may be the AOA antenna **730**.
(242) In certain embodiments, the one or more driving units **1911** and **1912** may not be included in the components of the electronic device **101**.
(243) The positioning circuitry **2110** may be a communication circuitry (e.g., the communication module **190**). The positioning circuitry **2110** may include an ultra-wide band (UWB) positioning circuitry **2111**. The positioning circuitry **2110** may transmit and receive a positioning signal from the first antenna **730** and/or the second antenna **2140**, and may transmit received signal information to the processor **120**.
(244) The processor **120** may determine the relative distance, direction, and/or altitude between the electronic device **101** and the external electronic device based on the signal information received from the positioning circuitry **2110**, and may control to display a sharing interface based on the determined relative distance, direction, and/or altitude between the electronic device **101** and the external electronic device on the display module **160**.
(245) The sensor circuitry **2120** may include at least one of a magnetic sensor (or a Hall sensor, **2121**) and/or an IR sensor (or an optical sensor, **2122**).
(246) The sensor circuitry **2120** may determine whether the display module **160** of the electronic device **101** is in a slit-in or slit-out state based on a change in the magnetic force using the magnetic sensor **2121** and/or a change in the detection pattern using the IR sensor **2122**.
(247) The near field communication circuitry **2130** may include at least one of a Bluetooth communication circuitry **2131** and/or a Wi-Fi communication circuitry **2132**. The near field communication circuitry **2130** may receive a message regarding communication establishment

(e.g., Bluetooth or Wi-Fi) with an external electronic device through the second antenna **2140** and/or the first antenna **730**, or may perform communication with the external electronic device communication.
(248) The second antenna **2140** may be an antenna formed on a side surface surrounding a space surrounding the front surface **701** and the rear surface **702** of the electronic device **101**. The second antenna **2140** may be an antenna included in the frame of the electronic device **101**. The second antenna **2140** is a directional antenna, and may receive a positioning message signal from the external electronic device and may transmit signal information to the processor **120**.
(249) In certain embodiments, when the electronic device **101** performs a sliding or rolling operation of the display module **160** using the driving units **1911** and **1912**, the sliding length of the display module **160** may be determined based on the number of rotations of the motor of the driving units and the rotation time thereof, thereby determining the open state, the closed state, and/or the intermediate state of the electronic device **101**. The electronic device according to certain embodiments disclosed herein may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smart phone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. The electronic device according to embodiments of the disclosure is not limited to those described above.
(250) It should be appreciated that certain embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or alternatives for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to designate similar or relevant elements. A singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “a first”, “a second”, “the first”, and “the second” may be used to simply distinguish a corresponding element from another, and does not limit the elements in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled/connected with/to the other element directly (e.g., wiredly), wirelessly, or via a third element.
(251) As used herein, the term “module” may include a unit implemented in hardware, software, or firmware, and may be interchangeably used with other terms, for example, “logic,” “logic block,” “component,” or “circuit”. The “module” may be a minimum unit of a single integrated component adapted to perform one or more functions, or a part thereof. For example, according to an embodiment, the “module” may be implemented in the form of an application-specific integrated circuit (ASIC).
(252) Certain embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., the internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a complier or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. The term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic

wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.
(253) According to an embodiment, a method according to certain embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., Play Store™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.
(254) According to certain embodiments, each element (e.g., a module or a program) of the above-described elements may include a single entity or multiple entities. According to certain embodiments, one or more of the above-described elements may be omitted, or one or more other elements may be added. Alternatively or additionally, a plurality of elements (e.g., modules or programs) may be integrated into a single element. In such a case, according to certain embodiments, the integrated element may still perform one or more functions of each of the plurality of elements in the same or similar manner as they are performed by a corresponding one of the plurality of elements before the integration. According to certain embodiments, operations performed by the module, the program, or another element may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

Claims

1. An electronic device, comprising: a display including a first screen area that is maintained in exposure to an external environment, and an extended screen area that is exposed or stowed, according to a state of the display of the electronic device; communication circuitry configured to transmit and receive a signal to or from an antenna disposed within the extended screen area, and wherein a location of the antenna changes in the electronic device changes when the state of the display of the electronic device changes; a memory; and a processor operatively coupled to the display, the antenna, the communication circuitry, and the memory, wherein the memory stores instructions executable by the processor to cause the electronic device to: determine a position of the electronic device relative to an external electronic device, using the antenna, execute a sharing operation with the external electronic device, identify whether the display of the electronic device is disposed in a first state or a second state, based on detecting that the display is disposed in the first state, perform a first positioning operation for the external electronic device, and display a user interface including first information based on the first positioning operation on the display.
2. The electronic device of claim 1, wherein the instructions are further executable by the processor to: based on detecting that the display is disposed in the second state, perform a second positioning operation for the external electronic device, and display the user interface including second information based on the second positioning operation on the display.
3. The electronic device of claim 1, wherein the instructions are further executable by the processor to: detect whether the state changes from the first state to the second state, based on detecting a change from the first state to the second state, perform a second positioning operation for the external electronic device, and control the display to display the user interface based on the second positioning operation.
4. The electronic device of claim 3, wherein the instructions are further executable by the processor to: simultaneously display the first information obtained during the first state and second information obtained during the second state for a predetermined time, when the state of changes

from the first state to the second state.
5. The electronic device of claim 4, wherein when the predetermined time lapses, the first information is removed from display, while the second information is maintained on display.
6. The electronic device of claim 1, wherein: the first state includes a closed state of the electronic device, and the second state includes an open state of the electronic device.
7. The electronic device of claim 6, wherein: in the open state, the extended screen area is extended from an interior of the electronic device to the external environment, via mechanically sliding out of the electronic device, and in the closed state, the extended screen area is stowed within the interior of the electronic device, via mechanically sliding into the electronic device.
8. The electronic device of claim 1, wherein the instructions are further executable by the processor to: display an image and/or an icon for the external electronic device corresponding to directivity of the antenna, and arrange the display of the image and/or the icon on the user interface to indicate a highest priority of the external electronic device for the sharing operation.
9. The electronic device of claim 1, further comprising: a near field communication circuitry, wherein the instructions are further executable by the processor to cause the electronic device to: determine whether a communication message is received from the external electronic device through the near field communication circuitry, determine whether the external electronic device is detected through the antenna when the communication message is received, and change the state of the electronic device when the external electronic device is not detected through the antenna.
10. The electronic device of claim 1, further comprising: a sensor circuitry, wherein the processor is configured to detect the state of the electronic device through the sensor circuitry based on at least one of a change in a magnetic force, and a change in a detection pattern.
11. A method in an electronic device, comprising: executing a sharing operation to transmit and receive data with an external electronic device using an antenna; detecting whether a display of the electronic device is in a first state or second state according to whether an extendable screen area of the display is stowed within the electronic device, or extended to be exposed to an external environment of the electronic device, and wherein a location of the antenna changes when the display of the electronic device from the first state to the second state; executing, by a processor, a first positioning operation for the external electronic device based on detecting the display is in the first state; and displaying a user interface including first information based on the first positioning operation on the display.
12. The method of claim 11, further comprising: based on detecting that the display is disposed in the second state, executing a second positioning operation for the external electronic device, and displaying the user interface including second information based on the second positioning operation on the display.
13. The method of claim 11, further comprising: detecting whether a state changes from the first state to the second state, based on detecting a change from the first state to the second state, executing a second positioning operation for the external electronic device, and displaying the user interface based on the second positioning operation.
14. The method of claim 13, further comprising: simultaneously displaying the first information obtained during the first state and second information obtained during the second state for a predetermined time, when the state of changes from the first state to the second state.
15. The method of claim 14, further comprising: when detecting elapse of the predetermined time, removing the first information obtained in the first state and displaying the second information obtained in the second state.
16. The method of claim 11, wherein: the first state includes a closed state of the electronic device, and the second state includes an open state of the electronic device.
17. The method of claim 16, wherein: in the open state, the extendable screen area is extended from an interior of the electronic device to the external environment, via mechanically sliding out of the electronic device, and in the closed state, the extendable screen area is stowed within the interior of

the electronic device, via mechanically sliding into the electronic device.

18. The method of claim 11, further comprising: displaying an image and/or an icon for the external electronic device corresponding to directivity of the antenna, and arranging the display of the image and/or the icon on the user interface to indicate a highest priority of the external electronic device for the sharing operation.

19. The method of claim 11, further comprising: determining whether a communication message is received through a near field communication (NFC) circuitry, determining whether the external electronic device having transmitted the communication message is detected through the antenna when the communication message is received, and changing a state of the electronic device when the external electronic device having transmitted the communication message is not detected through the antenna.

20. The method of claim 11, wherein s state of the display is detected based on at least one of a change in a magnetic force and a change in a detection pattern as detected through sensor circuitry of the electronic device.
